CampusIQ - Day 2 Notes

Name: Fahad Ahmed **Date:** _____

Topic: Databases, PostgreSQL and basic SQL

1. What is a Database?

- **Database** = a place where data is stored in an organised way. It is like Excel, but **much faster and safer.**
- Many people can use it at the same time, and the data stays safe.
- CampusIQ will keep student data (name, branch, CGPA) in a database.

Important words:

- **Table** = a sheet. It holds one kind of data (example: students).
- **Row** = one record. In our table, 1 row = 1 student.
- **Column** = one field (like name or cgpa).
- **SQL** = the language we use to ask questions to a database.
- **PostgreSQL** = the database software we installed.
- **pgAdmin** = the screen (tool) to use PostgreSQL easily.

2. Our students Table

This is how our table looks after we add the data:

roll_no	name	branch	cgpa
21AI01	Ravi	AIML	8.4
21AI02	Asha	AIML	7.2
21AI03	Imran	AIML	9.1
21AI04	Zoya	AIML	6.8

Top line = column names. Every other line = one row (one student).

3. Setup Steps

1. Install **PostgreSQL**. Remember the **password**. Keep port **5432**. (I got version 18, it works the same as 16.)
2. Open **pgAdmin 4**. Double-click the server and enter the password.
3. Right-click **Databases**, then Create, then Database. Name it **campus**, click Save.
4. Right-click **campus**, then click **Query Tool**.
5. Check the tab name. It must show **campus/postgres@PostgreSQL 18**. Then we are in the right database.
6. Paste the SQL in the editor and press **F5** to run it.

Write the password somewhere safe. We need it again and again.

4. CREATE TABLE

CREATE TABLE makes a new, empty table. Run it only once:

```
CREATE TABLE students (
  roll_no TEXT PRIMARY KEY,
  name TEXT,
  branch TEXT,
  cgpa NUMERIC(3,1)
);
```

Line by line:

- **CREATE TABLE students** = make a table named students.
- **roll_no TEXT PRIMARY KEY** = a text column. PRIMARY KEY means every row has a unique roll_no. It cannot repeat and cannot be empty.
- **name TEXT, branch TEXT** = more text columns.
- **cgpa NUMERIC(3,1)** = a number with 3 digits in total and 1 digit after the point. Example: 8.4. The biggest value is 99.9.
- The **semicolon** ; ends a SQL statement.

Data types: TEXT = words and letters. NUMERIC = numbers with decimals.

5. INSERT INTO

INSERT adds rows (data) into the table:

```
INSERT INTO students VALUES
('21AI01','Ravi','AIML',8.4),
('21AI02','Asha','AIML',7.2),
('21AI03','Imran','AIML',9.1),
('21AI04','Zoya','AIML',6.8);
```

- **INSERT INTO students** = add rows into the students table.
- **VALUES** = the data to add. The order must match the columns: roll_no, name, branch, cgpa.
- **Text goes in single quotes** like 'Ravi'. **Numbers have no quotes** like 8.4.
- Each row is inside brackets, rows are separated by commas, and the last row ends with ;
- Message **INSERT 0 4** means 4 rows were added.

6. SELECT - Asking Questions

SELECT reads data from the table. Run each query **one at a time.** Before the next query, click in the editor, press Ctrl+A and Delete.

Query 1: show everything

```
SELECT * FROM students;
```

The **star** * means all columns. Result: all 4 rows, as in our table.

Query 2: only students with CGPA above 8

```
SELECT name FROM students WHERE cgpa > 8;
```

WHERE filters the rows. Only rows where cgpa is above 8 are shown.

name
Ravi
Imran

Gufran's check: Ravi and Imran must come in the result.

Query 3: average CGPA

```
SELECT AVG(cgpa) FROM students;
```

AVG = average. (8.4 + 7.2 + 9.1 + 6.8) / 4 = 31.5 / 4 = **7.875**

pgAdmin may show it as 7.8750000000000000. It is the same number.

7. Extra Practice

Try these yourself. The expected result is written beside each one:

```
SELECT name, cgpa FROM students
WHERE cgpa < 7;
```

Result: **Zoya 6.8** (only one student is below 7).

```
SELECT * FROM students
ORDER BY cgpa DESC;
```

ORDER BY sorts the rows. **DESC** = high to low. Order: Imran, Ravi, Asha, Zoya.

```
SELECT COUNT(*) FROM students;
```

COUNT counts rows. Result: **4**.

```
SELECT MAX(cgpa) FROM students;
```

MAX gives the biggest value. Result: **9.1**. (MIN gives the smallest.)

Comparison signs:

```
=   equal to          <>  not equal to
>   greater than      <   less than
>=  greater or equal  <=  less or equal
```

8. Save the SQL and Push to GitHub

1. In VS Code make a folder **db** inside the **backend** folder.
2. Make a new file inside db and paste the CREATE, INSERT and 3 SELECT queries. Save with Ctrl+S. Full path and name:

```
backend/db/day2_students.sql
```

3. Open the terminal in the campusiq folder and run these one by one:

```
git status
git pull
git add .
git commit -m "day 2: students table sql"
git push
```

4. Open the GitHub repo and check that the SQL file is there, inside backend and then db.

Why save the SQL in a file? So that the team can see our work and run it again any time.

Repo link changed. It is now in the organization. One-time fix:

```
git remote set-url origin
  https://github.com/gufran-fahad-campusiq/campusiq.git
```

Notes images: save them with these names, then add, commit and push the same way:

```
notes/fahad/day01.png
notes/fahad/day02.png
```

9. Common Errors and Fixes

- **relation "students" already exists** - the table is already made. Do not run CREATE TABLE twice.
- **duplicate key value violates unique constraint** - that roll_no is already in the table. A PRIMARY KEY cannot repeat.
- **Only the last result shows** - pgAdmin shows only the last query. Run one at a time, or select one query with the mouse and press F5.
- **relation does not exist** - Query Tool is open on the wrong database. Open it from **campus**.
- **Syntax error** - check for a missing comma, a missing semicolon, or text without quotes.
- **Wrong password** - use the same password you set while installing PostgreSQL.
- **Server will not connect** - open Services in Windows, find postgresql-x64-18, right-click and Start.

DROP TABLE students; deletes the table with all its data. Use it only if you want to start again.

10. Revision Questions

Q1. What is the difference between a table, a row and a column?

A **table** is the whole sheet. A **row** is one student. A **column** is one field like name or cgpa.

Q2. What does WHERE cgpa > 8 do?

It filters the table and shows only the rows where cgpa is more than 8. Result: Ravi and Imran.

Q3. What is a PRIMARY KEY?

It is a column that gives every row a unique identity. It cannot repeat and cannot be empty. Here it is roll_no.

Q4. Difference between CREATE TABLE and INSERT?

CREATE TABLE makes the empty table (the structure). **INSERT** puts data (rows) inside it.

Q5. What does SELECT * mean?

It means show **all columns** from the table.

11. Day 2 Checklist

- ☐ PostgreSQL and pgAdmin are working
- ☐ Database campus is created
- ☐ Table students made with 4 rows
- ☐ 3 SELECT queries done, Ravi and Imran came for cgpa > 8
- ☐ SQL file saved in the backend and db folder
- ☐ Pushed to GitHub and file is visible
- ☐ Notes images added in my notes folder

Important: Practise SQL daily. Run one query at a time.